

Optimisation Model for Strategic Planning — Scheduling and Advertising in the TV Sector

Scheduling Optimization Model for Channel A

Abdulrahman Adeyinka Badmus

Abstract

This report provides an in-depth analysis of a strategic scheduling approach aimed at maximising viewership when advertisements are to be sold and thus increasing Channel A's revenue. It examines how the intelligent placement of advertisements and the careful scheduling of movies can significantly boost audience engagement. By considering viewer demographics and their peak viewing times, the programming schedule is optimised to attract a larger audience and maximise market share. The report proposes an optimised advertising strategy that can effectively complement our programming to generate higher viewership. Overall, this comprehensive approach is designed to ensure that Channel A remains competitive and in tune with the preferences of audiences.

Contents

1	Introduction	4
2	Literature Review	4
3	Problem Statement and Exploration	4
3.1	Movie database	5
3.2	Channel A's schedule	5
3.3	Competitors' schedules	5
3.4	Genre conversion rates for ad slots	6
4	General Model	6
4.1	Assumptions	6
4.2	Index Sets	6
4.3	Decision Variables	7
4.4	Parameters	7
4.5	Constraints	8
4.6	Objective Function	10
5	Implementation	10
6	Results	10
7	Sensitivity Analysis	11
8	Discussion, Challenges and Extension	12
8.1	Extensions and challenges	12
8.2	Assumptions Revisited and Final Consultancy Recommendations	14
8.2.1	Assumptions revisited	14
8.2.2	Consultancy Recommendations	14
9	Conclusions	15

List of Abbreviations

SPOT	Scheduling Programs Optimally for Television
MIP	Mixed Integer Programming
MVP	Minimum Viable Product
GPP	Gross Profit Percentage
HPC	High Performance Computing

1 Introduction

This report focuses on Channel A, a TV channel designed to simulate real-world challenges faced by broadcasters. Competing against three other channels in a market of 1 million viewers, Channel A aims to enhance its viewership and maximise revenue through intelligent scheduling and strategic advertising. To achieve this, the report proposes a Mixed-Integer Programming (MIP) model that optimises Channel A’s scheduling and advertising strategy. By aligning ad slot usage with viewership data, the model enables Channel A to increase its audience while maximising the value of its ad slots, directly linked to revenue generation. Furthermore, a sensitivity analysis is conducted to conclude an optimal budget for licensing fees and advertisement of the scheduled movies. The model and implementation in this report is an MVP, and several recommendations are provided to extend the given approach throughout the report.

Recent studies reveal a significant advantage in broadcasting channels with data-driven scheduling strategies, where movie scheduling decisions rely on detailed analyses of audience preferences, competitor behavior, and content performance metrics. There has been a significant shift in channels leveraging demographic insights, viewership baselines, and genre-based preferences to predict viewer behavior leading to an optimised movie schedule [2]. To this end, demographic preference and viewing habits are incorporated when scheduling movies. Furthermore, competitors’ schedules are considered when choosing ad slots, to maximise viewer conversions.

2 Literature Review

The optimisation of television scheduling and advertisement is extensively explored in operational research. This section reviews key contributions to the field with a focus on mathematical modelling techniques, demographic targeting, competitor analysis, and advertising strategies.

Mathematical modelling techniques such as MIP and heuristic algorithms are widely applied to solve complex scheduling problems. These models prove suitable as they balance multiple factors, including audience demographics, licensing costs, and advertising revenue potential, to provide an optimal schedule.

Reddy, for example, introduced the SPOT model, which uses a mixed-integer network flow methodology to optimise prime-time schedules by considering ratings, revenues, and licensing costs, and explicitly considered advertising revenues alongside program costs to maximise network profitability [4]. This model demonstrated superior flexibility by incorporating managerial inputs and constraints like lead-in effects and time slot restrictions. It also emphasises the effect of competitors’ schedules on an advertisement.

Furthermore, Danaher explored how conversion rates vary based on genre compatibility between advertised content and competitor programming [1]. For example, advertising a horror movie during a similar genre on a competitor channel yields higher conversion rates than advertising it during a romance-comedy slot.

Overall, these findings highlight the significance of genre-based targeting in optimising ad placements and reveal gaps in existing models. To address these gaps, the model incorporates the following feature: integration of competitors’ schedules — the proposed model incorporates competitor data to optimise ad placements and maximise audience conversion rates.

3 Problem Statement and Exploration

This report addresses the challenges of maximising Channel A’s viewership through a mathematically optimised, data-driven scheduling approach. The problem requires selecting movies that appeal to targeted demographics while considering competitors’ schedules, licensing and advertising costs, and ad placement strategy. The problem is cast as a MIP problem and solved

using the Xpress optimisation package in Python. The objective of this approach is to maximise viewership in time slots designated for selling ads.

Specifically, the model is implemented to maximise viewership when ads are sold, which in turn maximises revenue. Furthermore, the model accounts for competitors' schedules when selecting ad slots to advertise movies to target audiences most likely to watch the advertised movie based on genre similarity. The genres of each movie are considered when advertising; therefore only movies similar to those being shown on the competitor's channel are advertised in the same time slot.

Three demographics are considered significant in this problem: children, adults, and retirees. It is important to consider demographic preference when selecting movies and their viewing habits.

This problem involves four key data sets: a database of potential movies, Channel A's schedule, the schedules of three competitors, and each competitor's advertising conversion rates by genre. Below is a summary of the exploratory data analysis conducted and how these datasets were managed.

3.1 Movie database

This dataset contains 5,920 movies with the following relevant columns: genres, runtime with ads, scaled popularity for each demographic, and license fee. When running the model, several heuristics were used to reduce the size of the database due to computing power constraints.

To reduce the dataset, the movies were filtered with IMDb ratings between 5.5 and 7.5, released after 2000, and with licensing fees exceeding 1,000,000. This selection focused on moderate-quality, high-cost movies, representing the worst-case budget scenarios. Excluding unprofitable movies was avoided as long as they met these conditions, ensuring a comprehensive yet challenging dataset for the model.

This heuristic was developed based on the exploratory data analysis done in the `DataAnalysis_MovieDataBa` file on the project's GitHub repository [3]. The goal was to create a reduced dataset that emphasises high-budget constraints to test the model under tough conditions. By focusing on worst-case scenarios, the recommendations would remain valid even in less challenging situations. Segmenting based on popularity ensured optimal audience targeting.

A series of criteria were applied to the given movie dataset of 5,920 movies, reducing the dataset to 136 movies that meet the set criteria. Of these, 39 movies were chosen based on their popularity value for implementing the model.

3.2 Channel A's schedule

This dataset is structured as 5-minute time slots for a 12-week period, with relevant columns Date-Time and a baseline view count for each demographic. As the focus of this report is to schedule over a one-week period, the time slots beyond seven days were dropped. To reduce the size of the problem, the 5-minute time slots were aggregated into 30-minute time slots, averaging across the values of the columns when resampling. The Date-Time column then showed the starting time of each half-hour interval.

3.3 Competitors' schedules

Three competitors' schedules were provided, structured similarly, over 12 weeks in 5-minute time slots. They provided information on when a movie is shown or an ad slot is available to buy. To reduce problem size, these datasets were preprocessed to drop all non-advert 5-minute time slots. The relevant columns from this dataset are Date-Time, a baseline view count for each demographic, and ad slot price.

3.4 Genre conversion rates for ad slots

This dataset is provided for each channel and has columns for each possible genre. It provides a conversion rate for each ad slot available on that channel based on the movie that is being shown. If the movie has the genre, the conversion rate is non-zero, and is zero otherwise.

4 General Model

4.1 Assumptions

Several assumptions were made in designing the model, to reduce complexity of the initial MVP. In future, some of these assumptions could be further explored to improve the model's accuracy and robustness. Given more time and computing resources, stochastic elements and further constraints could be introduced to the model. The list of assumptions is laid out below:

- All ad slots can be sold at any time and the identity of the buyer is not a significant factor.
- There is no cost to selling ad slots, i.e., losing viewership if competitors advertise on Channel A.
- It is optimal to show each movie only once.
- The viewers gained from advertising a movie will watch the movie for its entire duration.
- Advertising a movie multiple times does not result in a diminishing number of viewers converted.
- A specific proportion of ad viewers are converted, i.e., converting viewers does not depend on some probabilistic element.
- The model cannot produce a scheduling and advertising strategy which can return a viewership greater than the total population, i.e., the total population distributed across the four channels is not tracked.
- The only revenue source for the channel is through the sale of advertisement slots.

4.2 Index Sets

- M , the set of candidate movies.
- T , the set of 30-minute time slots of our channel.
- G , the set of unique movie genres.
- C , the set of competitor channels.
- A_c , the set of ad slots available to buy on competitor channel c , $c \in C$.

4.3 Decision Variables

Binary decision variables:

$$\begin{aligned}
y_m &= \begin{cases} 1, & \text{if movie } m \text{ is to be shown, } m \in M. \\ 0, & \text{otherwise.} \end{cases} \\
x_{mt} &= \begin{cases} 1, & \text{if we show movie } m \text{ in time slot } t, m \in M, t \in T. \\ 0, & \text{otherwise.} \end{cases} \\
w_{mt} &= \begin{cases} 1, & \text{if we advertise movie } m \text{ in time slot } t \text{ on our own channel, } m \in M, t \in T. \\ 0, & \text{otherwise.} \end{cases} \\
v_{mt} &= \begin{cases} 1, & \text{if we sell the ad slot in time slot } t \text{ when movie } m \text{ is shown, } m \in M, t \in T. \\ 0, & \text{otherwise.} \end{cases} \\
z_{c,m,k_c} &= \begin{cases} 1, & \text{if we buy ad slot } k_c \text{ on channel } c \text{ to advertise movie } m, m \in M, k_c \in A_c, c \in C. \\ 0, & \text{otherwise.} \end{cases}
\end{aligned}$$

Continuous decision variables:

- u_{mt} , the total viewership gained from showing and advertising movie m in time slot t , $m \in M, t \in T$.
- q_{mt} , total viewership of movie m for time slot t if the ad slot is to be sold, $m \in M, t \in T$.
- s_m , the start time of movie m in minutes measured from 7am on the first day of the week, $m \in M$.
- e_m , the end time of movie m in minutes measured from 7am on the first day of the week, $m \in M$.

4.4 Parameters

- r_m , runtime of movie m in minutes, including ad breaks, $m \in M$.
- l , time slot length, 30 minutes in this problem.
- W , number of minutes in a week (10,080).
- B , total advertising and movie licensing budget (£). This value is explored using sensitivity analysis.
- α , the date-time of the first day of the week at 7am.
- α_t , the date-time of the start of time slot t , $t \in T$.
- β_{c,k_c} , the date-time when ad slot k_c is scheduled on channel c , $k_c \in A_c, c \in C$.
- P , the total population of viewers across our channel and the competitor channels; $P = 1,000,000$ in this problem.
- p_{md} , the scaled popularity of movie m for demographic d , $m \in M, d \in D$.
- b_{dt} , the baseline view count of time slot t for demographic d , $d \in D, t \in T$.
- $\tilde{b}_{c,d,k_c}$, the baseline view count of ad slot k_c , for channel c , for demographic d , $d \in D, k_c \in A_c, c \in C$.

- κ_{mt} , the conversion rate for advertising movie m on time slot t , $m \in M$, $t \in T$.
- $\tilde{\kappa}_{c,g,k_c}$, the conversion rate for advertising on ad slot k_c on channel c if the movie has genre g , $c \in C$, $k_c \in A_c$, $g \in G$.
- δ_{mg} , whether movie m has genre g , computed before running the model:

$$\delta_{mg} = \begin{cases} 1, & \text{if movie } m \text{ has genre } g, m \in M, g \in G. \\ 0, & \text{otherwise.} \end{cases}$$

- f_i , licensing fee for movie i , $i \in M$.
- a_{k_c} , the cost of buying ad slot k_c on competitor channel c , $c \in C$, $k_c \in A_c$.

The parameters r_m , β_{c,k_c} , α_t , β_{dt} , r_m , $\tilde{b}_{c,d,k_c}$, b_{dt} , p_{md} , a_{k_c} , f_i , $\tilde{\kappa}_{c,g,k_c}$ were derived from the datasets provided.

The parameter κ_{mt} (conversion rate for advertising movie m on time slot t) was calculated when the model is run, using the formula:

$$\kappa_{mt} = \frac{\sum_{d \in D} p_{md} b_{dt}}{\sum_{d \in D} b_{dt}} \quad (1)$$

4.5 Constraints

Only one movie can be scheduled for each half-hour time slot:

$$\sum_{m \in M} x_{mt} = 1, \quad \forall t \in T \quad (2)$$

A movie is only scheduled if it is to be shown:

$$x_{mt} \leq y_m, \quad \forall m \in M, t \in T \quad (3)$$

Movies must be scheduled for their whole runtime:

$$l \sum_{t \in T} x_{mt} = y_m r_m, \quad \forall m \in M \quad (4)$$

The start time and end time of a movie must differ by the length of the movie. This constraint also ensures that movies are not scheduled over the broadcasting gap between 12am and 7am, and is linked with Constraints (5) and (6) to force that time slots for which the movie is scheduled are consecutive:

$$e_m - s_m = y_m r_m, \quad \forall m \in M \quad (5)$$

These constraints ensure that the movie is scheduled only for time slots which are consecutive. Along with Constraint (4), this forces that the time slots which the movie is scheduled for must fall within a gap that is the length of the movie. The parameter W ensures that s_m is not set to zero when the movie is not scheduled at time slot t :

$$e_m \geq x_{mt}(\alpha_t + l - \alpha), \quad \forall m \in M, \forall t \in T \quad (6)$$

$$s_m \leq x_{mt}(\alpha_t - \alpha) + W(1 - x_{mt}), \quad \forall m \in M, \forall t \in T \quad (7)$$

Only one ad can be bought per available ad slot on competitor channels:

$$\sum_{m \in M} z_{c,m,k_c} \leq 1, \quad \forall k_c \in A_c, \forall c \in C \quad (8)$$

Constraint (8) ensures that for each time slot on our channel only one ad slot is shown. Either the ad slot is sold, or our own movie is advertised. This constraint, along with Constraint (16), controls whether advertisements are sold rather than advertising our own movie:

$$\sum_{m \in M} w_{mt} + \sum_{m \in M} v_{mt} = 1, \quad \forall t \in T \quad (9)$$

The movie is only advertised on our channel or on competitor channels if it is to be shown:

$$z_{c,m,k_c} \leq y_m, \quad \forall m \in M, \forall k_c \in A_c, \forall c \in C \quad (10)$$

$$w_{mt} \leq y_m, \quad \forall m \in M, \forall t \in T \quad (11)$$

Ads are only sold when a movie is scheduled, if the movie is scheduled:

$$v_{mt} \leq x_{mt}, \quad \forall m \in M, t \in T \quad (12)$$

Ads are bought to advertise a movie only before the first time slot the movie is scheduled to be shown:

$$z_{c,m,k_c}(\beta_{c,k_c} + l - \alpha) \leq s_m, \quad \forall m \in M, \forall k_c \in A_c, \forall c \in C \quad (13)$$

$$w_{mt}(\alpha_t + l - \alpha) \leq s_m, \quad \forall m \in M, \forall t \in T \quad (14)$$

Constraint (14) is used to count the total viewership gained from showing and advertising a movie in each time slot it is scheduled. This constraint uses the assumption that any viewers converted by advertising watch the movie for each time slot the movie is scheduled. The first sum gets the base viewership gained from scheduling the movie in time slot t , calculated as the sum of the popularity times the baseline viewership for each demographic. The second sum calculates the total viewership gained from advertising the movie on competitor channels, calculated as the sum of the popularity times the baseline viewership times a conversion rate for each genre of the movie being shown on the competitor channel at the time of the ad slot; each term is multiplied by a genre factor δ_{mg} , which ensures viewers are only converted when movie m and the movie showing on the competitor channel share genres. The third sum counts the viewership gained from advertising movie m on our own channel, similar to the first sum, with an added conversion rate:

$$\begin{aligned} u_{mt} \leq & P \sum_{d \in D} p_{md} b_{dt} x_{mt} \\ & + P \sum_{k_c \in A_c} \sum_{g \in G} \sum_{d \in D} p_{md} \tilde{b}_{c,d,k_c} \tilde{\kappa}_{c,m,g,k_c} \delta_{mg} z_{c,m,k_c} \\ & + P \sum_{t \in T} \sum_{d \in D} p_{md} b_{dt} \kappa_{mt} w_{mt}, \\ & \forall m \in M, \forall t \in T, \forall c \in C \end{aligned} \quad (15)$$

The cost of licensed movies and ad slots bought must be less than or equal to the total budget allocated:

$$\sum_{m \in M} y_m f_m + \sum_{m \in M} \sum_{c \in C} \sum_{k_c \in A_c} z_{c,m,k_c} \leq B \quad (16)$$

Constraint (16) ensures the viewership gained from showing and advertising a movie in a time slot is only counted in the objective function if an ad is sold in that time slot. This ensures that the amount of viewers gained from advertising a movie which is to be shown later is greater than the amount of viewers in the current time slot:

$$u_{mt} \leq P v_{mt}, \quad m \in M, t \in T \quad (17)$$

4.6 Objective Function

In the objective function, viewership for time slots in which ad slots are sold is maximised. The viewership for each time slot is tracked, which allows the calculation of total revenue after the model finishes solving.

$$\max \left(\sum_{m \in M} \sum_{t \in T} u_{mt} \right) \quad (18)$$

5 Implementation

Given the complexity of the problem, solving the model directly runs into runtime issues due to time frame and computing resource limitations. Instead, to illustrate the effectiveness of this modelling strategy, the model is run over a three-day block for a reduced dataset of movies. This movie set was selected using the methodology laid out in Section 3.1. The budget constraint is also not used in this initial run of the model, allowing us to get an idea of what an optimal schedule might cost over a three-day period.

This implementation demonstrates that the model does in fact schedule movies and advertises effectively, and extension to seven days is a matter of computing resources.

Despite these limitations, this approach allows us to conduct a sensitivity analysis on the best value to set for the budget. Furthermore, it provides a useful illustration of our method in smaller time frames and emphasises the fact that the model can easily be extended to a week-long interval, given the appropriate resources.

6 Results

After running the model for a three-day period (5/10/2024–7/10/2024), the average viewership gained across slots is 73,670. Following [5], we propose selling ad slots on a Cost Per Thousand (CPT) basis. After rounding the viewership in each time slot designated to be sold down to the nearest thousand, we get 73,000. The total cost of this scheduling and advertising strategy is £29,143,421. The gap between best bound and objective value for this solution was 82.76%. These values enable the calculation of Gross Profit Percentage (GPP), based on a range of selling prices. Candidate selling prices and their corresponding GPP are shown in Table 1. A further analysis of Channel A’s operating cost and the current market would be required to select an optimal selling price.

Table 1: CPT selling prices and their corresponding GPP

CPT Selling Price	1800	1820	1840	1860	1880	1900	1920	1940	1960	1980	2000
GPP	1.755	2.835	3.891	4.924	5.936	6.926	7.895	8.845	9.775	10.686	11.579

The model proposes that 45% of available ad slots on our channel are sold, while 55% are used to advertise movies that will be shown at a later point. This shows that the model does not naively sell all available ad slots, but instead carefully selects which slots should be used to advertise in order to maximise viewership for those which will be sold.

Unfortunately, further analysis of these figures highlights some limitations of the model. For instance, on 7/10/2024, “The Amazing Spider-Man” is scheduled to be shown from 21:30 until 00:00. It is advertised on our channel a total of 15 times and is not advertised on competitor channels. The model suggests that all 1,000,000 potential viewers in the population tune in to watch the movie in each time slot the movie is scheduled. This value reflects a poor choice of conversion rate for advertising on our own channel, as well as the limiting assumption that

there is no diminishing return on viewers converted when advertising the same movie multiple times on the same channel.

Despite there being 667 available slots for purchase on competitor channels and no budget constraint used in this case, the model only selects 20 competitor slots to use for advertising. This highlights the model’s careful handling of ad slot purchases even before considering cost. Potential slots are only considered if the movie to be advertised has enough genres similar to the movie being shown when the ad slot is available. This way, only viewers interested in movies similar to those advertised are targeted.

7 Sensitivity Analysis

The sensitivity analysis was conducted to explore how varying budget constraints influence the model’s outcomes, including scheduling decisions, advertising strategies, and viewership. The approach taken began with an initial unconstrained run of the model as described in Sections 5 and 6, observing costs without budget limitations. This provided valuable insights into the resource requirements and helped define realistic budget intervals for further analysis.

For this analysis, we tested the budgets: 30,100,000; 30,050,000; 30,000,000; 29,100,000; 29,050,000; 29,000,000; and 28,500,000. Through this approach, the analysis aimed to highlight key aspects such as the impact of reduced budgets on viewership, the adaptability of advertising strategies, and the overall cost-effectiveness of different schedules.

Table 2: Sensitivity analysis results

Budget	Cost	Average Viewership	Cost/Viewership Ratio
No Limit	29,143,421.70	73,669.80	395.52
30,100,000	29,774,392.85	104,078.17	286.11
30,050,000	29,415,486.02	151,729.48	193.86
30,000,000	28,990,269.37	105,957.46	273.54
29,100,000	28,393,074.05	103,340.50	274.74
29,050,000	28,352,577.53	106,511.57	266.16
29,000,000	28,731,757.56	137,430.05	209.03
28,500,000	27,561,679.41	140,366.91	196.31

Table 2 shows that a budget of 30,050,000 offers the best cost-to-viewership ratio of 193.86, indicating optimal efficiency. Higher ratios for other budgets, including the “No Limit” budget, which is the worst, show a 395.52 ratio, indicating less efficient use of resources.

Let us now consider a sensitivity analysis studying the relationship between gross profit and selling price for different budget scenarios. The curves show almost the same behaviour for all budgets, with an almost linear relationship between gross profit and selling price, accompanied by a small parabolic curvature. The main difference lies in the attainable gross profit for each budget. Interestingly, the highest gross profit is achieved with a budget of 30,050,000.

We can also observe that the smallest gross profit growth with price occurs when there is no budget limit. This is coherent with our intuition; when there is no constraint on the budget, the system can become unoptimised, leading to less efficient allocation of resources and diminishing returns on gross profit growth as the price increases.

Let us finally consider the ads. Figure 2 summarises the distribution of ads placed on the own channel, ads sold, and ads bought from competitors across different budgets. It shows how the advertising strategy varies with budget changes, highlighting the trends for each budget level.



Figure 1: Gross profit vs. selling price for all the budgets

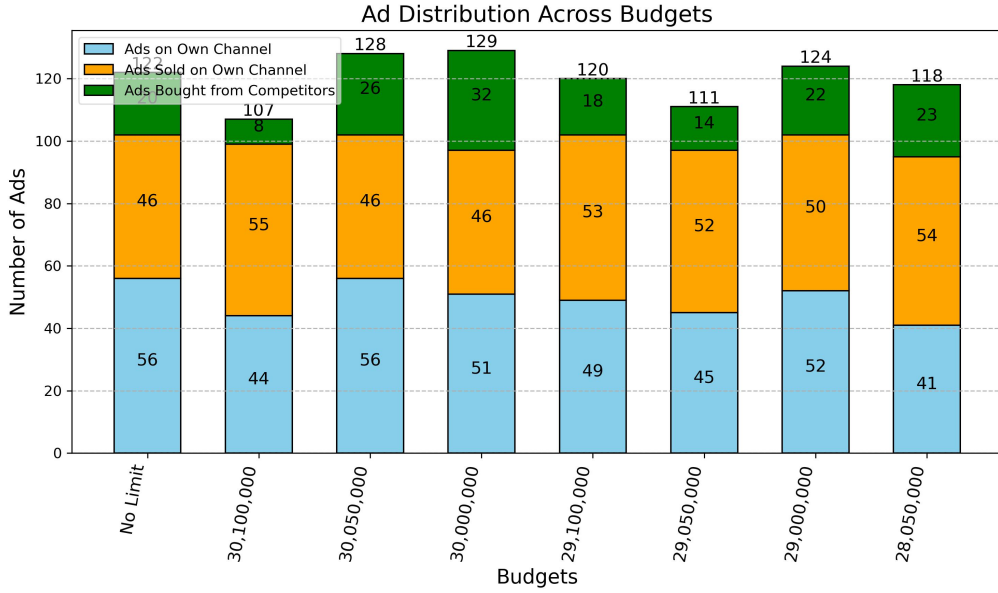


Figure 2: Ad distribution across budgets

8 Discussion, Challenges and Extension

8.1 Extensions and challenges

Given the time complexity of the problem, solving the model directly ran into runtime issues due to time frame and computing resource limitations. Our MVP approach aimed to produce a schedule over a three-day block using the reduced movie dataset discussed in Section 3.1. The plan was to run the model for the first four-day block, then for the second three-day block.

However, our implementation demonstrates the model’s ability to balance demographic preferences, advertising costs, and competitor actions effectively. Despite computational constraints,

the framework is implemented in a highly generalisable way, capable of handling a seven-day scheduling horizon and accommodating a much larger set of movies. It is designed to scale seamlessly with sufficient resources. Although out of scope for an MVP approach, here we give an extensive explanation of limitations, and what could be done if there were no resource limitations.

Firstly, the model should be extended to a seven-day period and more movies should be reintroduced into the dataset. This would allow a more comprehensive exploration of which movies are beneficial to schedule based on our demographics’ viewing habits.

Secondly, the model agreed on for this final report counted viewers deterministically, as noted in the assumptions section, assuming all potential viewers from advertising would be captured. This choice was made to reduce computational cost and problem complexity while ensuring reasonable results. However, we did attempt a probabilistic approach, where actual viewers were sampled based on conversion rate probabilities for a given time slot. The total conversion probability p was computed as a scalar product between the conversion rate vector at an advertisement time slot and the conversion of the movie to be shown, accounting for the fact that there must be a match between the movie to advertise and the movie shown during the advertisement, with the standard deviation of the normal distribution set to $\max(0, (1 - p) \cdot \mu)$, where μ was the expected viewers. This ensured higher precision when $p \rightarrow 1$, while maintaining variability for lower probabilities. If sampled viewers exceeded μ , values were adjusted to lie uniformly within the upper quartile of μ , ensuring realistic bounds. While this approach succeeded in capturing variability, it was computationally expensive, preventing extensive testing within the project’s timeline. Interested readers can refer to the file `probabilities_integrated.1.py` on the project’s GitHub repository [3] for implementation details and potential future extensions.

Finally, to extend the model beyond seven days adaptively, as suggested by the problem statement, two approaches were considered. The first, which we could call a “naive” approach but still reasonable, would be to run the model on the newly released schedule of other channels. This would be acceptable within the framework of the project, as inherently there is no real game-theory-like competition. In fact, our channel has a significant advantage, as it has access to the schedules from other channels at the beginning of the week and can adapt its schedule accordingly to maximise profit, while the other channels are unaware of what we plan. Thus, the game is skewed in our favour, and the other channels cannot be considered rational or optimal players. This is why this naive approach could make sense in this specific context.

The second approach would be to adjust the prices of our advertisement slots and decide which ad slots to disable based on the potential viewers we might lose if we sell the slot, compared to the number of viewers converted from our channel by a specific channel in previous runs (which can be considered a measure of how competitive that channel is). The price would vary depending on the channel, and ad slots would be disabled if the potential loss in viewers exceeds a certain threshold. This threshold could be updated iteratively based on the global ratio between the viewers converted from our channel previously and the maximum number of viewers that could have been converted from our channel. This would quantify the general market competitiveness, with the threshold being adjusted over time by analysing data from previous weeks. The key aspect here is that both the number of viewers and the conversion rates would be updated dynamically, allowing the model to “learn” from past weeks’ performance and adapt its strategy for future decisions, simply by deciding whether to disable a slot and by deciding the prices of slots for each viewership and channel. This learning process would enable the model to continuously improve its decision-making in response to evolving competitive dynamics.

8.2 Assumptions Revisited and Final Consultancy Recommendations

8.2.1 Assumptions revisited

Revisiting the assumptions laid out above, here are some recommendations as to how they could be addressed:

- The model should consider whether ad slots are being sold to competitors and consider an associated cost in the objective function for viewers who are lost to competitor advertisements.
- When a movie is advertised there should be a dampening effect on the number of viewers converted as the number of advertisements used increases, especially for repeated advertisements on the same channel. This would reflect the diminishing return of viewers converted and the same viewers being targeted repeatedly.
- As our intention is to sell ad slots for a price on a CPT basis, it would be more optimal to cast u as an integer decision variable. Then the left-hand side of Constraint (14) would be written as $1000u$ and the right-hand side of Constraint (16) as $\frac{P}{1000}v_{mt}$. The objective function would look similar but would now maximise every 1,000 units of viewers. This approach would better handle edge cases than our current implementation, i.e., ensure that adverts on competitor channels are only bought when they increase viewership by at least another 1,000 viewers. This is not implemented, as casting u as an integer variable greatly increases the problem complexity.

8.2.2 Consultancy Recommendations

After conducting this study, we now have 10 consultancy recommendations and comments:

1. **Budget Allocation:** We recommend allocating £60M–£70M for a 7-day period, scaling from the successful £30.05M used in the 3-day model. This ensures an optimal cost-to-viewership ratio below 200.
2. **Computational Resources:** We recommend investing in cloud or HPC solutions to handle extended 7-day scheduling and larger datasets, leveraging parallel computing and advanced optimisation solvers.
3. **Adaptive Adjustments:** We recommend starting with weekly strategy updates based on competitor data and gradually transitioning to probabilistic modelling using conversion rates and stochastic scenarios.
4. **Targeted Advertising:** We recommend focusing on high-conversion competitor ad slots for similar genres and prime-time self-advertising, ensuring alignment with demographic preferences.
5. **Incremental Improvements:** We recommend implementing dynamic ad pricing, genre-specific targeting, and repeated ad dampening effects to optimise conversions while efficiently utilising resources.
6. **Sensitivity Testing:** We recommend performing sensitivity tests with 10–15% budget adjustments to refine resource allocation strategies while maintaining a cost-to-viewership ratio below 200.
7. **Optimized Ad Slot Management:** We recommend selling high-value slots during peak times, buying competitor slots with high conversion rates, and prioritising multi-genre movies to maximise audience impact.

8. **Extensions for Future Improvements:** We recommend introducing a 7-day adaptive model, refining stochastic viewer sampling, and adjusting pricing thresholds dynamically.
9. **Operational Efficiency:** We recommend setting minimum revenue thresholds to prevent bankruptcy and adjusting operational resources in line with demographic trends and market demands.
10. **Long-Term Strategies:** We recommend analysing audience demographics to expand, for example, children’s programming, exploring shorter movies or classics, and using stochastic simulations to refine strategies over a 12-week horizon.

9 Conclusions

This report proposes an approach that can produce an effective and data-driven scheduling and advertising strategy for Channel A, which maximises viewership in time slots in which the channel sells advertisements. The model chooses carefully whether to use ad slots for advertising Channel A’s movies or to sell these ad slots. Furthermore, the model intelligently selects which competitor ad slots should be bought, considering the competitor’s schedule and the movie to be advertised. That is, ad slots are chosen based on movie similarity to the movie being shown, ensuring advertisements are specific and targeted. Current results produced by the model are for three-day blocks, but extending to seven days is simply a matter of making more computing resources available.

The full implementation and code for this project can be accessed on the project’s GitHub repository [3].

References

- [1] P. J. Danaher and M. T. Mawhinney. *Optimizing television program schedules using choice modeling*. Marketing Science, 2001.
- [2] FilmGrail. *Data-driven showtime optimization for cinemas*, 2024. <https://filmgrail.com/blog/data-driven-showtime-optimization-for-cinemas>.
- [3] Group9-MMCS-Edinburgh. *MMCS Project Repository*. https://github.com/DavidKrame/MMCS_Project_UoE_Group9, 2024. Accessed: 2024-11-25.
- [4] M. K. Reddy et al. *SPOT: Scheduling Programs Optimally for Television*. Management Science, 1998.
- [5] TV Advertising Agency. *The cost of TV advertising*, 2024. <https://tvadvertising.agency/the-cost-of-tv-advertising/>.